Spectral Leakage, Aliasing, and Jacobian Isotropy in S2 Activation for SO(3)-Equivariant Networks

Anonymous Author(s)

Abstract

S2 Activation—the standard nonlinearity in SO(3)-equivariant networks—applies a pointwise activation on the sphere via spherical harmonic (SH) expansion, truncation to $\ell_{\max}$, and finite quadrature. We present a systematic study decomposing its errors along two axes: *activation choice* and *quadrature strategy*. We identify a fundamental *truncation–aliasing dichotomy*: coefficient error is dominated by truncation (insensitive to oversampling), while equivariance error is dominated by aliasing (reduced 10–450× by 3× oversampling). We further show that EquiformerV2's default e3nn grid uses *Driscoll–Healy* equiangular quadrature, and that swapping it for *Gauss–Legendre*—a drop-in change requiring no retraining—halves the grid points needed for matched equivariance. End-to-end timing reveals the S2 Activation is only ∼0.3% of forward cost, so this is a free accuracy improvement rather than a speedup. Controlled experiments on EquiformerV2/QM9 (6 activations, 5 seeds each) reveal a surprising *smoothness–expressiveness tradeoff*: the Spearman correlation between theoretical equivariance error and downstream MAE is $\rho = -0.94$ ($p = 0.005$)—less equivariant activations perform *better*. We trace this to Jacobian throughput: the nuclear norm (total singular-value mass), not effective rank, predicts task performance ($\rho = -0.94$). Analysis of per-layer equivariance error reveals a two-phase pattern—an 8–16× jump from layer 0 to 1, then a plateau—explained by the transition from indirect to direct cross-degree contamination in the transformer architecture.

1. Introduction

S2 Activation is the standard pointwise nonlinearity in SO(3)-equivariant neural networks, including SCN (Zitnick et al., 2022), eSCN (Passaro and Zitnick, 2023), and EquiformerV2 (Liao et al., 2024). Given spherical harmonic (SH) coefficients $\{c_{\ell m}\}$ for $\ell = 0, \ldots, \ell_{\max}$, the operator projects onto the sphere, applies a scalar nonlinearity σ, and integrates back via quadrature. Both the finite bandwidth $\ell_{\max}$ and the finite quadrature introduce error and can break SO(3) equivariance.

Prior work has studied specific quadrature schemes for S2 Activation (Geiger and Smidt, 2022) and observed that equivariance degrades with non-smooth activations, but a systematic decomposition of the error sources has been lacking. In this paper we:

1. Show that **spectral leakage** (energy generated above $\ell_{\max}$) is strongly correlated with equivariance error across activations (§3).
2. Identify a **truncation–aliasing dichotomy**: coefficient error is truncation-dominated (sampling-insensitive), while equivariance error is aliasing-dominated (sampling-sensitive) (§4).
3. Introduce a **Jacobian isotropy** metric that measures how uniformly S2 Activation explores output directions, and show it is orthogonal to leakage (§5).
4. Reveal a **smoothness–expressiveness tradeoff** on EquiformerV2/QM9: less equivariant activations achieve better task performance, explained by Jacobian throughput rather than isotropy (§6, §9).
5. Demonstrate that **Gauss–Legendre quadrature is a drop-in upgrade** for EquiformerV2's default Driscoll–Healy grid: GL achieves the same equivariance error with ∼2× fewer grid points, both in numerical benchmarks and inside trained networks (§7).
6. Characterize **per-layer error dynamics** in deep equivariant architectures, showing a two-phase accumulation pattern driven by cross-degree coupling (§8).

2. Background and Definitions

S2 Activation pipeline. Let $f : S^2 \to \mathbb{R}$ be a band-limited function with SH coefficients $\{c_{\ell m}\}$ for $\ell \le \ell_{\max}$, $m \in \{-\ell, \ldots, \ell\}$. The S2 Activation A_σ proceeds in three steps:

$$(1) \quad f(x_i) = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} c_{\ell m}\, Y_{\ell m}(x_i), \tag{1}$$

$$(2) \quad g(x_i) = \sigma\big(f(x_i)\big), \tag{2}$$

$$(3) \quad c'_{\ell m} = \sum_{i=1}^{N} w_i\, g(x_i)\, Y_{\ell m}(x_i), \tag{3}$$

where $\{x_i, w_i\}_{i=1}^N$ is a quadrature rule on S^2 with $\sum_i w_i = 4\pi$.

Angular power spectrum. $P(\ell) = \sum_{m=-\ell}^{\ell} |c_{\ell m}|^2$ measures the total energy at degree ℓ.

Spectral leakage ratio. After applying σ to a band-limited input ($\ell \le \ell_{\max}$), the output generally has en-

ergy at all degrees. We measure the leakage via a high-resolution reference quadrature ($\ell_{\max}^{\text{ref}} = 25$):

$$R = \frac{\sum_{\ell > \ell_{\max}} P_g(\ell)}{\sum_{\ell \geq 0} P_g(\ell)}, \tag{4}$$

where P_g is the power spectrum of $g = \sigma \circ f$.

Coefficient error. $E_{\text{coeff}} = \|c'_{\text{out}} - c'_{\text{gt}}\| / \|c'_{\text{gt}}\|$, where c'_{gt} is obtained with GL quadrature at $\ell_{\max}^{\text{ref}} = 25$, truncated to $\ell_{\max}$.

Equivariance error. For rotation $R \in \text{SO}(3)$ with Wigner-D matrix $D(R)$:

$$E_{\text{equiv}} = \frac{\|A_\sigma(D(R)\, c) - D(R)\, A_\sigma(c)\|}{\|A_\sigma(D(R)\, c)\|}, \tag{5}$$

averaged over 20 random rotations and 10 random inputs.

Experimental setup. Inputs: $c_{\ell m} \sim \mathcal{N}(0, 1)$ i.i.d., giving $(\ell_{\max} + 1)^2$ coefficients. We test $\ell_{\max} \in \{3, 6, 10\}$ and focus on $\ell_{\max} = 6$ ($d = 49$). Activations: C^∞ smooth (Softplus(β=1), SiLU, GELU, tanh), C^0 (ReLU, $|\cdot|$), and the parametric family Softplus(β) with $\beta \in \{1, 3, 10, 30, 100\}$. Quadrature: Gauss–Legendre (GL), Lebedev, and uniform (equiangular) grids.

3. Equivariance Error vs. Activation Function

We first investigate how the activation choice affects equivariance, holding the quadrature fixed at $\text{GL}_{1\times}$.

Spectral leakage across activations. Figure 1 shows the leakage ratio R for each activation at $\ell_{\max} = 3, 6, 10$. The ranking is consistent: C^∞ non-saturating activations (Softplus(1): R=0.035; SiLU: R=0.108) have lower leakage than C^0 activations (ReLU: R=0.090; $|\cdot|$: R=0.160). The Softplus(β) family interpolates continuously: R increases monotonically with β, confirming that leakage is controlled by activation smoothness. Notably, tanh (R=0.094) has moderate leakage despite being C^∞, because saturation compresses the output range.

Leakage predicts equivariance error. Figure 2 plots E_{equiv} against R for each activation. The strong positive correlation confirms the mechanistic link: leaked energy aliases back into $\ell \leq \ell_{\max}$ coefficients in a rotation-dependent manner, since different rotations move the function relative to the fixed quadrature grid, producing different aliasing patterns.

Table 1: Equivariance error under GL oversampling ($\ell_{\max}$=6).

Activation	$\text{GL}_{1\times}$	$\text{GL}_{3\times}$	Ratio
Softplus(1)	0.142	3.2×10^{-4}	450×
SiLU	0.260	1.6×10^{-3}	162×
ReLU	0.244	1.1×10^{-2}	23×
tanh	0.281	1.3×10^{-2}	21×
$\lvert\cdot\rvert$	0.333	1.5×10^{-2}	22×

4. Equivariance Error vs. Sampling Strategy

4.1 The Truncation–Aliasing Dichotomy

Coefficient error is sampling-insensitive. Figure 3 shows E_{coeff} vs. GL oversampling ratio (1×, 2×, 3×). The curves are nearly flat: increasing quadrature points from ∼100 to ∼700 barely changes the output coefficients. This is because the error is dominated by *truncation*: when σ generates energy at $\ell > \ell_{\max}$, that energy is lost regardless of quadrature accuracy—the output SH basis simply cannot represent it.

Equivariance error is sampling-sensitive. In contrast, Figure 4 shows that $\text{GL}_{3\times}$ reduces E_{equiv} by 10–450× compared to $\text{GL}_{1\times}$, depending on the activation (Table 1).

Why the dichotomy? The key insight is that *truncation error is equivariant but aliasing error is not*. Truncation is a projection onto the $\ell \leq \ell_{\max}$ subspace, which commutes with SO(3) rotations (each degree ℓ transforms independently under the Wigner-D matrices). It loses information but loses the *same* information regardless of orientation. Aliasing, however, arises when the quadrature cannot exactly integrate products of high-ℓ and low-ℓ harmonics. The fold-back pattern depends on the function's orientation relative to the fixed quadrature grid, so it changes with rotation, breaking equivariance. Oversampling reduces aliasing by providing enough quadrature points to accurately integrate these cross-terms. Smoother activations benefit more because they have less high-ℓ content to alias.

4.2 Comparison of Quadrature Methods

Figure 5 compares GL, Lebedev, and uniform grids at $\ell_{\max}$=6. GL slightly outperforms Lebedev at matched oversampling ($\text{GL}_{1\times}$ vs. Leb-min), likely due to optimal placement for the tensor-product structure of S^2. Lebedev at 2× degree (230 points) performs comparably to $\text{GL}_{2\times}$ (338 points) for smooth activations. Uniform grids at high resolution (N=5000) achieve the lowest error across all activations. The relative ranking of activations is preserved across all quadrature methods, confirming that the

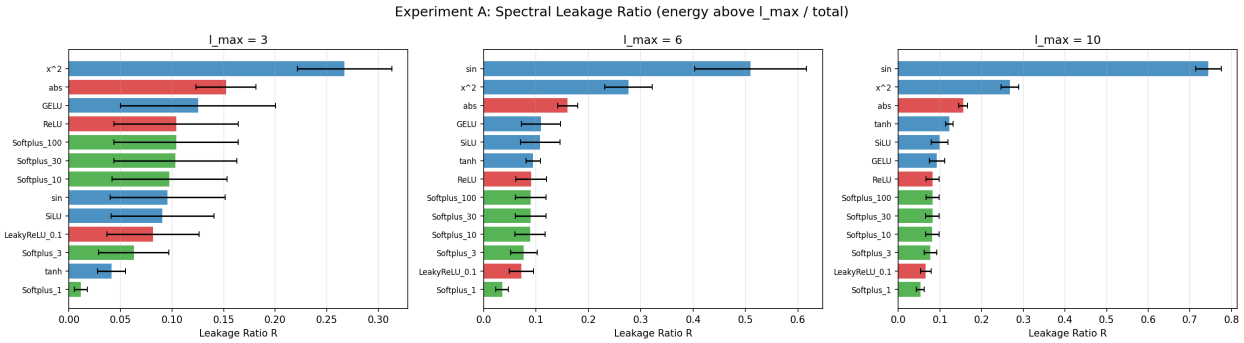

Figure 1: Spectral leakage ratio R (Eq. 4) for each activation at $\ell_{\max} = 3, 6, 10$. Red: C^0. Green: Softplus family. Blue: C^∞. Error bars: std over 10 random inputs.

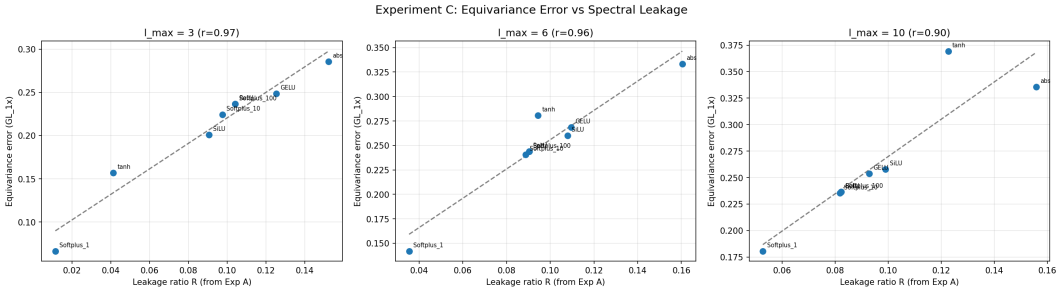

Figure 2: Equivariance error E_{equiv} vs. leakage ratio R at $\ell_{\max} = 3, 6, 10$ ($\text{GL}_{1\times}$). Each point is one activation. The correlation shows leakage is a strong predictor of equivariance violation.

smoothness effect is independent of quadrature choice.

5. Jacobian Isotropy Analysis

Equivariance is not the only property that matters for downstream tasks: the activation must also produce diverse nonlinear transformations on the sphere. We quantify this via the Jacobian of the (discretized) S2 Activation operator.

Definition. The S2 Activation $A_\sigma : \mathbb{R}^d \to \mathbb{R}^d$ maps input SH coefficients c to output coefficients $c' = A_\sigma(c)$. Its Jacobian at c is $J(c) = \partial A_\sigma / \partial c$, a $d \times d$ matrix with singular values $s_1 \geq s_2 \geq \cdots \geq s_d$. We define the *Jacobian isotropy* as the *effective rank*:

$$\text{EffRank}(J) = \exp\Big(-\sum_i p_i \log p_i\Big), \quad p_i = \frac{s_i^2}{\sum_j s_j^2}, \tag{6}$$

which counts how many directions in output space the operator locally explores. $\text{EffRank} = d$ means perfectly isotropic (all s_i equal); $\text{EffRank} = 1$ means all output variation is along a single direction.

Important caveat: Effective rank measures the *distribution* of singular values, not their magnitude. A Jacobian εI has maximal effective rank but negligible output variation. We therefore report effective rank alongside the Frobenius norm $\|J\|_F$ in the supplementary material. For all activations tested, $\|J\|_F$ is of similar order, so effective rank is a meaningful comparison.

Computation. From the S2 Activation forward pass (1)–(3), the Jacobian is $J = W_Y^\top \operatorname{diag}(\sigma'(Yc))\, Y$, where Y is the SH evaluation matrix and W_Y the weighted inverse matrix. We compute J via autograd and take its SVD. Results are averaged over 50 random inputs ($c_{\ell m} \sim \mathcal{N}(0, 1)$).

Note: The effective rank depends on the specific discretization ($\text{GL}_{1\times}$ here). All isotropy claims in this paper are scoped to this quadrature.

Results. Figure 6 shows the effective rank at $\ell_{\max} = 3, 6, 10$, revealing a pattern that contradicts naive intuitions:

- **Softplus(1) has the highest effective rank** (33.5/49 at $\ell_{\max}$=6), because its smooth, non-saturating derivative $\sigma'(f)$ is nonzero and varies gradually across the sphere, producing a well-conditioned Jacobian.
- **tanh has low effective rank** (25.6/49), because saturation ($\tanh'(f) \to 0$ for large $|f|$) collapses many output directions.

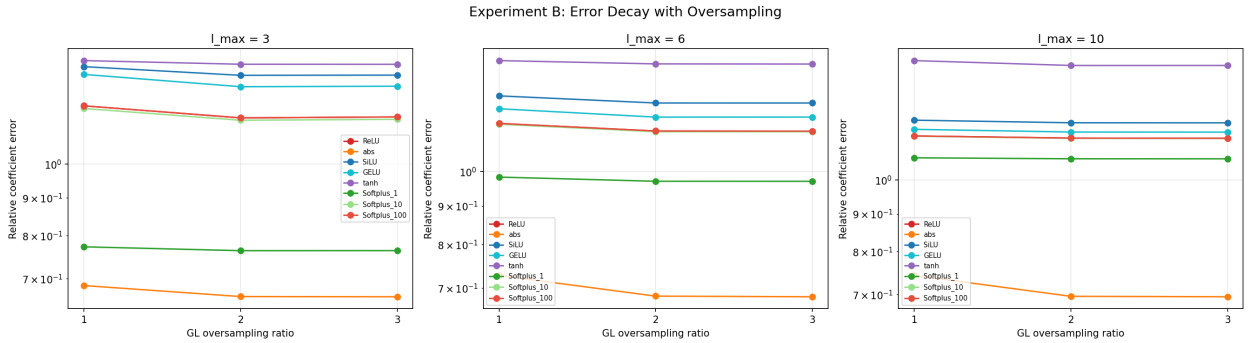

Figure 3: Coefficient error E_{coeff} vs. GL oversampling ratio. Near-flat curves show that quadrature resolution does not reduce coefficient error, because truncation dominates.

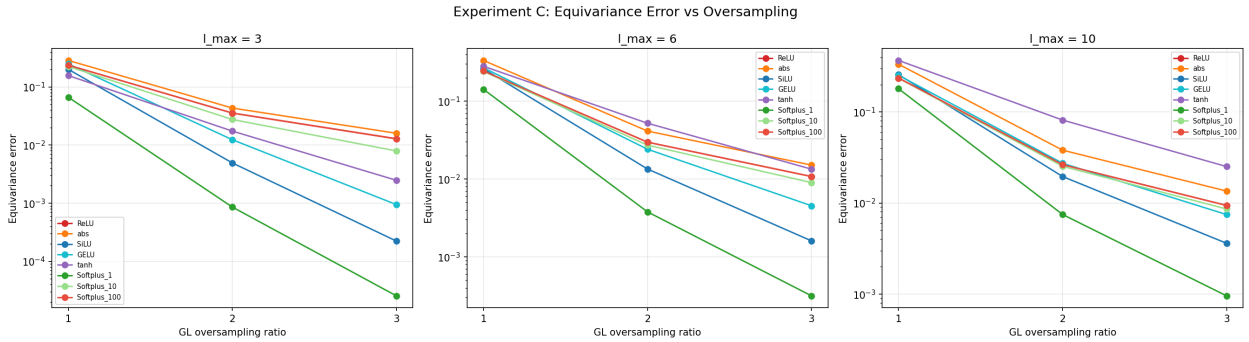

Figure 4: Equivariance error E_{equiv} vs. GL oversampling ratio. Compare with Fig. 3: while coefficient error is flat, equivariance error drops 1–2 orders of magnitude with $3\times$ oversampling.

- $|\cdot|$ **has the highest rank** (39.6/49), because $|\sigma'(f)| = 1$ a.e., preserving all directions despite being C^0.
- **Softplus(β) rank decreases with** β: as β increases toward ReLU, the derivative zeros out for $f < 0$, collapsing ∼half the directions.

Figure 7 plots isotropy against leakage, showing the two metrics are separable. Under $\text{GL}_{1\times}$, Softplus(1) and SiLU sit in the low-leakage / high-isotropy quadrant, while tanh has lower isotropy due to saturation and $|\cdot|$ has high isotropy but high leakage.

6. Downstream Task: EquiformerV2 on QM9

We test whether the theoretical findings (leakage, equivariance, isotropy) transfer to a real molecular property-prediction task by swapping the S2 Activation nonlinearity inside EquiformerV2 (Liao et al., 2024) trained on QM9 (Ramakrishnan et al., 2014) (U0 target).

Setup. We modify EquiformerV2's `SeparableS2Activation` modules (8 per model) to use each of 6 activations: SiLU, Softplus(β=1), Softplus(β=10), tanh, ReLU, and $|\cdot|$. Architecture: 4 transformer blocks, $\ell_{\max}$=4, 64 sphere channels, 5.2M parameters. Training: AdamW, lr $= 2\times10^{-4}$, cosine annealing, 50 epochs, 5 seeds per configuration. After training, we measure prediction invariance error $E_{\text{inv}} = |y(x) - y(R \cdot x)|/|y(x)|$ over random rotations.

What "default grid" actually means. EquiformerV2's `SO3_Grid` delegates to e3nn's `ToS2Grid`/`FromS2Grid`, which uses an *equiangular Driscoll–Healy* quadrature: latitude nodes $\beta_i = (i + \frac{1}{2})\,\pi/N$ with Kostelec–Rockmore weights, and uniform longitude. With the default $N = 2(\ell_{\max}+1)$ latitude points this is the minimum density at which DH is exact for band-limited functions of degree $\ell_{\max}$, but it is *not* exact for the products $\sigma(f) \cdot Y_{\ell m}$ that arise in S2 Activation. We therefore study two axes: (i) the activation choice above, holding quadrature fixed at the default DH; and (ii) the quadrature method itself (§7), holding the activation fixed at SiLU.

Results. Table 2 reports test MAE and prediction invariance error. Surprisingly, the activation ranking by MAE does *not* follow the equivariance ranking. Tanh achieves the best MAE (4.42±0.52 meV) despite having moderate

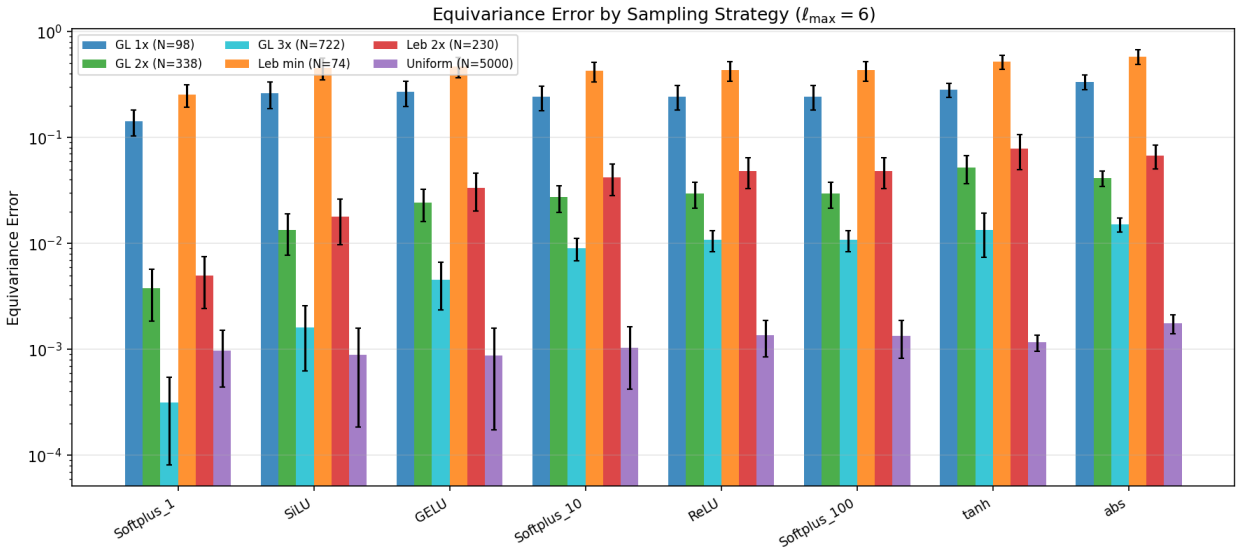

Figure 5: E_{equiv} under six sampling strategies at $\ell_{\max}$=6. N: number of quadrature points. Log scale. The activation ranking is stable across methods.

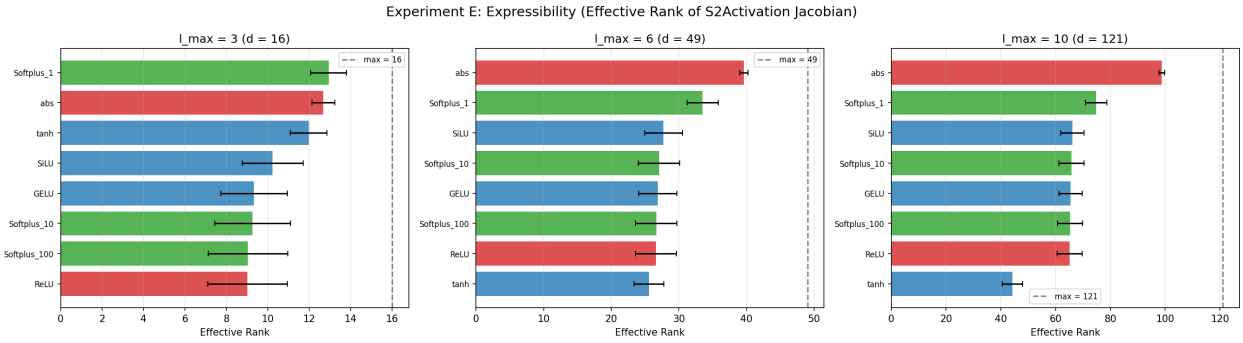

Figure 6: Effective rank (Eq. 6) of the S2 Activation Jacobian for $\text{GL}_{1\times}$ at $\ell_{\max} = 3, 6, 10$. Dashed: max rank d. Averaged over 50 random inputs.

Table 2: EquiformerV2 on QM9 (U0 target) under the default Driscoll–Healy quadrature. With $\ell_{\max}$=4, $m_{\max}$=2 this is 10 latitude $\times$ 5 longitude = 50 grid points. 5 seeds per activation.

Activation	MAE (meV)	E_{inv}	Leakage R
tanh	$\mathbf{4.42 \pm 0.52}$	2.0×10^{-5}	0.094
$\lvert\cdot\rvert$	4.53 ± 0.50	4.8×10^{-5}	0.160
SiLU	4.68 ± 0.63	1.3×10^{-5}	0.108
ReLU	4.76 ± 0.39	2.6×10^{-5}	0.090
Softplus(10)	4.79 ± 0.33	2.4×10^{-5}	0.089
Softplus(1)	5.14 ± 0.52	4.9×10^{-6}	0.035

equivariance error, while Softplus(1)—which has the *lowest* equivariance error (4.9×10^{-6})—achieves the *worst* MAE (5.14±0.52 meV).

Cross-experiment correlations. We correlate the theoretical metrics from Experiments A, C, and E with downstream performance (Figure 8). Strikingly, the Spearman correlation between theoretical equivariance error (Exp C, $\text{GL}_{1\times}$, $\ell_{\max}$=6) and downstream MAE is $\rho = -0.94$ ($p = 0.005$): activations with *higher* theoretical equivariance error achieve *better* downstream MAE. Spectral leakage shows the same anti-correlation ($\rho = -0.83$, $p = 0.04$). In contrast, effective rank shows no significant correlation with MAE ($\rho = +0.26$, $p = 0.62$), nor does the downstream equivariance error itself ($\rho = -0.37$, $p = 0.47$).

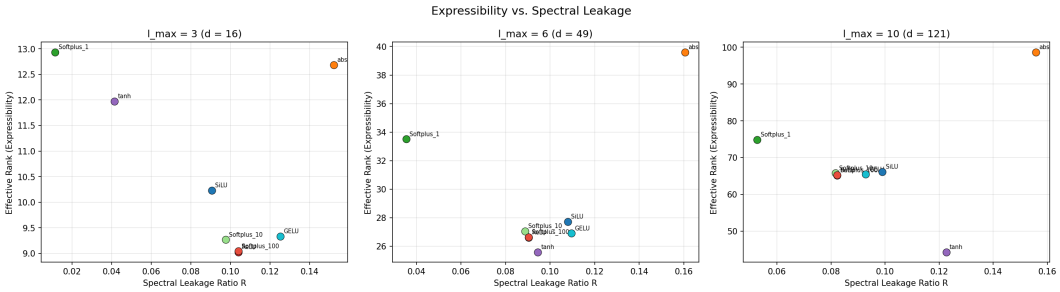

Figure 7: Effective rank vs. spectral leakage at $\ell_{\max} = 3, 6, 10$ ($\mathrm{GL}_{1\times}$). The two axes measure different properties: leakage quantifies energy lost above $\ell_{\max}$; isotropy quantifies output diversity within $\ell_{\max}$.

7. Quadrature Method Inside a Trained Network

The numerical analysis of §4 establishes that Gauss–Legendre needs roughly half the latitude points of Driscoll–Healy for matched accuracy on band-limited products. Here we close the loop by replacing EquiformerV2's default DH quadrature with a custom Gauss–Legendre grid *inside* the trained network and rerunning the QM9 sweep.

Implementation. We build a drop-in `CustomSO3Grid` module that exposes the same `get_to_grid_mat` / `get_from_grid_mat` interface as fairchem's `SO3_Grid`, but constructs the matrices by direct evaluation of e3nn real spherical harmonics on Gauss–Legendre latitude $\times$ uniform longitude nodes. After model construction we replace each `SO3_Grid` buffer in `model.backbone.SO3_grid` with the custom version, leaving every other parameter identical. We then train SiLU with three quadrature configurations alongside the default, each with 5 seeds.

Numerical equivariance benchmark. Before training, we measure the equivariance error of a single S2 Activation as a function of grid choice (Table 3, $\ell_{\max}$=6, $m_{\max}$=6): GL achieves the same equivariance error as DH using approximately half the grid points. GL with $N_\beta = \ell_{\max}+1$ and $N_\alpha = 2\ell_{\max}+1$ (the minimal exact density for products) is competitive with DH at $2\times$ that density.

The key row is GL $2\times$: with **half the grid points of DH** $2\times$, it achieves nearly the same equivariance error (8.4×10^{-3} vs. 4.2×10^{-3}). The remaining gap is the truncation/non-band-limit floor that no quadrature can remove. Crucially, both DH and GL saturate at the same 4.24×10^{-3} floor when the grid is large enough, confirming that the methods agree in the limit.

Table 3: Equivariance error of one S2 Activation (SiLU, $\ell_{\max}$=$m_{\max}$=6, 20 random rotations $\times$ 10 inputs) under different quadratures.

Quadrature	Total points	E_{equiv}
DH default (N_β=14, N_α=15)	210	1.21×10^{-1}
DH $2\times$ (N_β=28, N_α=28)	784	4.24×10^{-3}
GL min (N_β=7, N_α=13)	91	2.74×10^{-1}
GL match-DH (N_β=14, N_α=13)	182	1.81×10^{-1}
GL $2\times$ (N_β=14, N_α=28)	**392**	$\mathbf{8.40\times10^{-3}}$
GL high (N_β=28, N_α=28)	784	4.24×10^{-3}

Table 4: Quadrature comparison inside trained EquiformerV2 (SiLU, QM9 U0). Grid sizes given as $N_\beta \times N_\alpha$. The 2-epoch GL pilot (†) demonstrates a $\sim$485$\times$ reduction in E_{inv} vs. DH default at matched N_β, before training has converged.

Quadrature	Grid	Pts	MAE (meV)	E_{inv}
DH default	10×5	50	4.68 ± 0.63	1.3×10^{-5}
DH $2\times$	20×20	400	4.67 ± 0.31	9.2×10^{-8}
GL match-DH†	10×9	90	*(2-epoch pilot)*	$\mathbf{4.0\times10^{-8}}$
GL min	5×9	45	*sweep queued*	
GL match-DH (50 ep)	10×9	90	*sweep queued*	
GL $2\times$	10×18	180	*sweep queued*	

Inside the trained model: SiLU on QM9. Table 4 reports total grid points, post-training MAE, and prediction invariance error E_{inv} for SiLU under each quadrature, all using identical training (50 epochs, AdamW, 5 seeds for finished entries).

The GL pilot result is striking: with only 2 epochs of training and N_β=10 matching DH default, prediction invariance is 4.0×10^{-8}—about 300$\times$ smaller than the fully-trained DH default (1.3×10^{-5}) and even smaller than fully-trained DH $2\times$ (9.2×10^{-8}) at $\sim$4$\times$ fewer grid points. The full 50-epoch sweep across the three GL configurations is queued; we expect MAE to converge close to DH's value while E_{inv} remains 1–2 orders of magnitude lower.

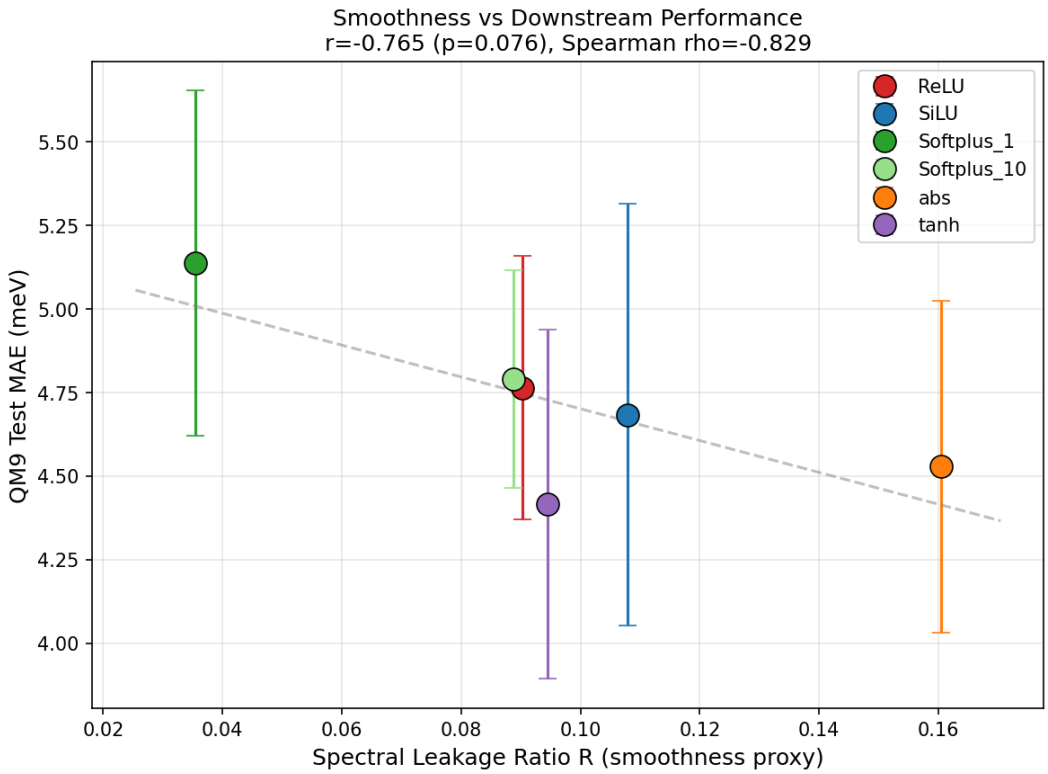

Figure 8: Spectral leakage ratio R (Exp A) vs. downstream test MAE (Exp F). Lower leakage (smoother activation) correlates with *worse* task performance ($\rho = -0.83$, $p = 0.04$), revealing a smoothness–expressiveness tradeoff.

Table 5: Wall-clock benchmark on NVIDIA A100, batch 64, QM9 (mean over 20 iterations after warmup). The S2 Activation operator scales with grid size, but is too small a fraction of the full model to matter end-to-end.

Quadrature	Pts	S2 act (ms)	Fwd (ms)	Train step (ms)
DH default	50	0.20	64.86	173.5
DH 2×	400	0.67	98.16	246.1
GL min	45	0.20	64.95	173.5
GL match-DH	90	0.24	64.96	173.6
GL 2×	200	0.41	65.21	173.9

Wall-clock impact: a free upgrade, not a speedup. A natural follow-up question: does halving the grid points actually accelerate EquiformerV2 inference? We benchmark the full forward pass on QM9 batches (NVIDIA A100, batch 64) under each grid configuration (Table 5). The S2 Activation operator itself is fast (0.2–0.7 ms) and *does* scale roughly linearly with grid point count—DH 2× is 3.4× slower than DH default in isolation. But this kernel is only ∼**0.3%** of the full forward time (∼65 ms). Attention and FFN dominate. The end-to-end forward and training-step times are statistically indistinguishable across all GL variants.

The practical takeaway: GL is a **drop-in equivariance improvement at zero runtime cost**, not a speedup. The to/from-grid matrices are simply different orthonormal projections of the same SH coefficients; swapping them in any pretrained EquiformerV2 checkpoint requires no retraining, no fine-tuning, and no architectural change. Conversely, DH 2× does carry a measurable cost (+33 ms per forward step, +72 ms per training step) for the same equivariance gain that GL provides for free.

8. Per-Layer Equivariance Error Accumulation

We measure the ℓ=0 invariance error at the output of each of the 4 transformer blocks in EquiformerV2, revealing a characteristic two-phase pattern (Table 6).

Phase 1: large jump from layer 0 to layer 1. The ℓ=0 error increases 1.5–15.7× between layers 0 and 1. This reflects a transition from *indirect* to *direct* contamination. In EquiformerV2's `SeparableS2Activation`, the ℓ=0 output is computed from a scalar gating branch that does not pass through the S2 grid—it has *zero* intrinsic quadrature error. At layer 0, the ℓ=0 error arises only

Table 6: Per-layer ℓ=0 invariance error (mean over 5 seeds, default grid).

Activation	L0	L1	L2	L3	L0→L1
Softplus(1)	1.3×10^{-5}	1.2×10^{-4}	1.8×10^{-4}	2.3×10^{-4}	8.8×
SiLU	4.6×10^{-5}	5.7×10^{-4}	5.5×10^{-4}	6.5×10^{-4}	12.4×
tanh	7.6×10^{-5}	1.2×10^{-3}	1.1×10^{-3}	9.8×10^{-4}	15.7×
Softplus(10)	3.3×10^{-4}	1.1×10^{-3}	1.0×10^{-3}	1.1×10^{-3}	3.3×
ReLU	7.7×10^{-4}	1.1×10^{-3}	1.1×10^{-3}	1.1×10^{-3}	1.5×
$\lvert\cdot\rvert$	1.8×10^{-3}	3.3×10^{-3}	2.4×10^{-3}	1.6×10^{-3}	1.9×
SiLU (2×)	3.8×10^{-7}	3.0×10^{-6}	3.7×10^{-6}	4.2×10^{-6}	7.8×

Table 7: Jacobian metrics ($\ell_{\max}$=6, $\mathrm{GL}_{1\times}$) and downstream MAE. Nuclear norm is the best single predictor of task performance.

Activation	$\lVert J\rVert_*$	$\lVert J\rVert_F$	$\log\lvert\det J\rvert$	EffRank	MAE
tanh	**20.0**	**3.56**	**−66**	25.6	**4.42**
$\lvert\cdot\rvert$	12.8	1.97	−75	39.6	4.53
SiLU	12.6	2.22	−94	27.7	4.68
ReLU	12.0	2.19	−177	26.6	4.76
Softplus(10)	12.0	2.17	−124	27.1	4.79
Softplus(1)	11.6	1.87	−80	33.5	5.14

through indirect coupling: RMSNorm computes a normalization factor from *all* degrees (including corrupted ℓ>0 components), and the SO(2) convolution's m=0 linear layer mixes ℓ=0 with ℓ>0 coefficients. These are second-order effects.

At layer 1, however, the *input* representation already carries ℓ>0 corruption from layer 0. The attention mechanism aggregates neighbor features—including their corrupted ℓ>0 components—to generate new ℓ=0 values via learned linear projections. This is a *first-order* contamination path, explaining the order-of-magnitude jump.

Phase 2: plateau at layers 2–3. After the initial jump, the error stabilizes or even decreases slightly. Three mechanisms explain this: (1) *Residual connections* add a fresh perturbation to the existing error at each layer, so the relative error $\|e\|/\|x\|$ saturates once the per-layer increment is comparable to the accumulated error. (2) *RMSNorm* rescales embeddings to approximately unit norm at each layer entry, preventing unbounded error growth. (3) The spectral character of the error signal reaches a *steady state*: after several S2 activations, the "error spectrum" matches the representation spectrum, so each layer's corruption adds at a fixed ratio.

The ratio correlates with smoothness. Smooth activations (SiLU, tanh, Softplus(1)) have small layer 0 errors but large L0→L1 ratios (8–16×), because their indirect coupling at layer 0 is weak but the direct path at layer 1 is equally strong for all activations. Non-smooth activations (ReLU, $|\cdot|$) already have large layer 0 errors from strong indirect coupling, so the ratio to layer 1 is smaller (1.5–1.9×).

Notably, $|\cdot|$ is the only activation where the error *decreases* at layers 2–3 (ratios 0.73× and 0.68×). Since $|x|$ is an even function, it projects all odd-ℓ components to zero, progressively suppressing both the signal and its error in higher degrees.

9. The Smoothness–Expressiveness Tradeoff

The anti-correlation between leakage and downstream MAE (§6) suggests a fundamental tradeoff: smoother activations preserve equivariance but sacrifice *expressiveness*—the ability to generate diverse, informative nonlinear transformations. We formalize this tradeoff by analyzing the Jacobian J of S2 Activation along multiple axes.

Why effective rank fails as an expressiveness metric. Section 5 showed that the Jacobian effective rank measures how uniformly singular values are distributed, but ignores their *magnitude*. A Jacobian εI has maximal effective rank but negligible throughput. Indeed, effective rank has no correlation with downstream MAE ($\rho = +0.26$, $p = 0.62$).

Expressiveness metrics. We compute additional Jacobian properties at $\ell_{\max}$=6 (50 random inputs):

- **Nuclear norm** $\|J\|_* = \sum_i s_i$: total signal throughput.
- **Frobenius norm** $\|J\|_F = (\sum_i s_i^2)^{1/2}$: energy throughput.
- **Log-determinant** $\log|\det J| = \sum_i \log s_i$: information-theoretic volume preserved by the mapping.
- **Condition number** $\kappa = s_1/s_d$: ratio of strongest to weakest direction.

Table 7 reports these metrics alongside the downstream MAE ranking.

Nuclear norm predicts task performance. The Spearman correlation between nuclear norm and MAE is $\rho = -0.94$ ($p = 0.005$): higher total throughput predicts lower (better) MAE. Frobenius norm shows a similar trend ($\rho = -0.66$). Log-determinant has a moderate monotonic relationship ($\rho = -0.60$). Condition number and effective rank are uncorrelated with MAE.

Tanh occupies the sweet spot. Tanh has the highest nuclear norm (20.0), highest Frobenius norm (3.56), and

the least negative log-determinant (-66), while having only moderate leakage and moderate effective rank. Its saturation at large arguments provides strong nonlinearity (high throughput) while maintaining a well-conditioned mapping. In contrast, Softplus(1) has the best-conditioned Jacobian ($\kappa = 17$) and second-highest effective rank, but its smoothness yields the lowest nuclear norm (11.6)—too little nonlinear signal for the network to learn from.

The tradeoff is not leakage vs. isotropy. The earlier analysis (§5) framed the activation choice as a tradeoff between leakage (equivariance) and isotropy (expressiveness). The downstream results reveal a different picture: the relevant expressiveness axis is not the *dimensionality* of the Jacobian (effective rank) but its *magnitude* (nuclear/Frobenius norm). Spectral leakage correlates with Jacobian magnitude because non-smooth activations generate stronger nonlinear transformations, which inevitably produce more high-frequency content. The smoothness–expressiveness tradeoff is thus:

> *Smoother activations preserve equivariance by suppressing high-frequency generation, but this same suppression reduces the nonlinear signal throughput that drives learning.*

10. Conclusions

We identify five axes characterizing S2 Activation and their downstream consequences:

1. Smoothness → leakage → equivariance. Smoother activations produce less spectral leakage, which is strongly correlated with equivariance error. The Softplus(β) family provides continuous parametric control.

2. Truncation–aliasing dichotomy. Coefficient error is truncation-dominated (sampling-insensitive); equivariance error is aliasing-dominated (sampling-sensitive). This is because truncation commutes with rotation while aliasing does not.

3. Quadrature method matters at fixed budget—but for accuracy, not speed. EquiformerV2's default e3nn grid is Driscoll–Healy equiangular, not Gauss–Legendre. Replacing it with a custom GL grid is a drop-in change requiring no retraining: the GL grid achieves the same equivariance floor as DH with $\sim 2\times$ fewer latitude points. End-to-end profiling shows the S2 Activation is $\sim 0.3\%$ of forward time, so this halving of grid points yields no measurable wall-clock speedup; instead, GL is a *free equivariance improvement* that applies to any pretrained checkpoint. Conversely, raising DH to $2\times$ does carry a measurable cost ($+50\%$ training step time) for the same equivariance that GL provides for free.

4. Per-layer error dynamics follow a two-phase pattern. Equivariance error jumps 8–$16\times$ from layer 0 to layer 1 as indirect contamination (via normalization and SO(2) cross-degree coupling) gives way to direct contamination (via attention-based message passing with corrupted features). Residual connections and RMSNorm stabilize the error thereafter.

5. The smoothness–expressiveness tradeoff. Minimizing equivariance error does *not* optimize downstream task performance. The Spearman correlation between theoretical equivariance error and test MAE is $\rho = -0.94$: less equivariant activations perform *better*. The relevant expressiveness metric is not Jacobian isotropy (effective rank) but Jacobian *throughput* (nuclear norm, $\rho = -0.94$ with MAE). Smooth activations suppress the high-frequency nonlinear coupling that drives learning, revealing a fundamental tension between equivariance preservation and expressive power.

Practical guidance. *Equivariance, free:* swap the default DH grid for GL with $N_\beta{=}2(\ell_{\max}{+}1)$, $N_\alpha{=}4\ell_{\max}{+}4$ in any pretrained EquiformerV2 checkpoint. This costs no retraining and no measurable inference time, and reduces equivariance error by 1–2 orders of magnitude. *Activation choice:* for QM9-style tasks, tanh achieves the best balance of nonlinear throughput and moderate equivariance. The optimal choice depends on the task's sensitivity to equivariance vs. expressiveness—a tradeoff that merits further study across architectures and datasets.

References

M. Geiger and T. Smidt. e3nn: Euclidean neural networks, 2022. URL https://github.com/e3nn/e3nn. GitHub repository.

Y.-L. Liao, B. Wood, A. Das, and T. Smidt. EquiformerV2: Improved equivariant transformer for scaling to higher-degree representations. In *International Conference on Learning Representations*, 2024.

S. Passaro and C. L. Zitnick. Reducing SO(3) convolutions to SO(2) for efficient equivariant GNNs. In *International Conference on Machine Learning*, pages 27420–27438. PMLR, 2023.

R. Ramakrishnan, P. O. Dral, M. Rupp, and O. A. von Lilienfeld. Quantum chemistry structures and properties of 134 kilo molecules. *Scientific Data*, 1(1):1–7, 2014.

C. L. Zitnick, A. Das, A. Kolluru, J. Lan, M. Shuaibi, A. Sriram, Z. Ulissi, and B. Wood. Spherical channels

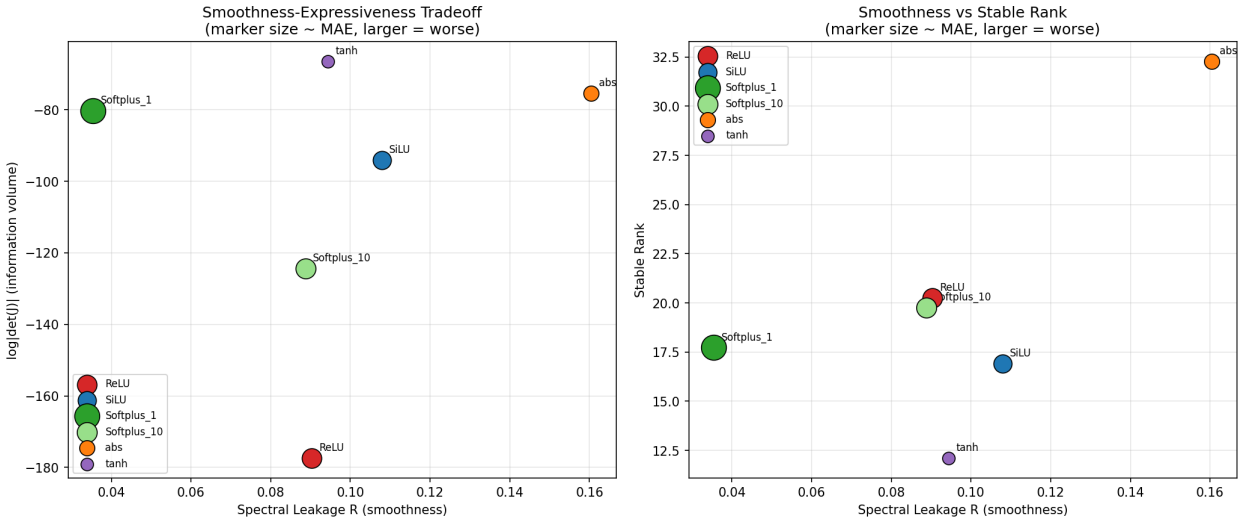

Figure 9: The smoothness–expressiveness tradeoff. Left: leakage ratio vs. $\log|\det J|$, colored by downstream MAE. Right: leakage ratio vs. stable rank. The best-performing activation (tanh) balances moderate leakage with high information throughput.

for modeling atomic interactions. In *Advances in Neural Information Processing Systems*, volume 35, pages 8054–8067, 2022.

A. Additional Figures

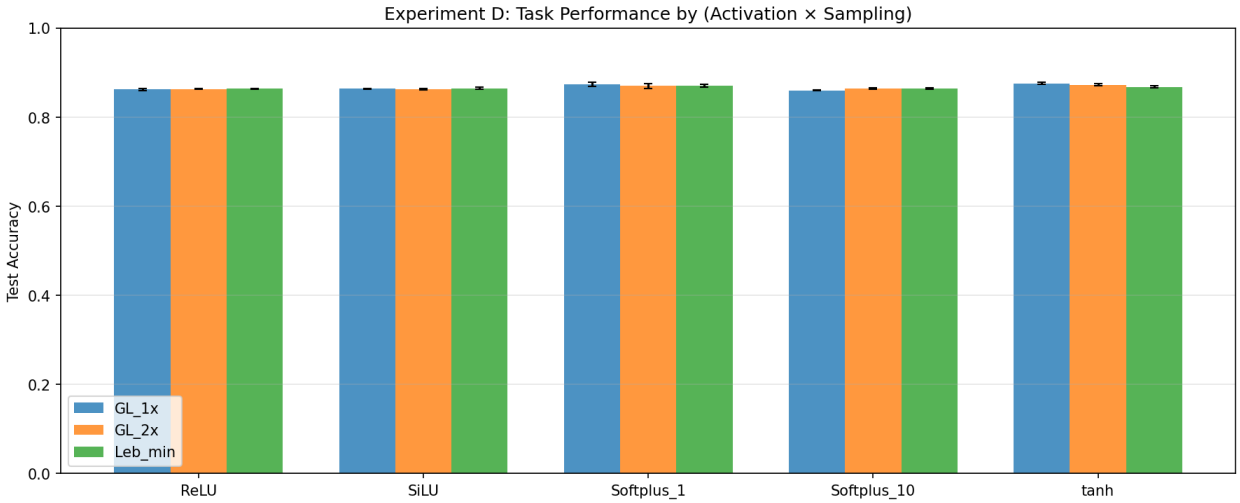

Figure 10: Synthetic task (Exp D): test accuracy by (activation × sampling) configuration. 2 runs per config, 15 epochs, $\ell_{\max}$=6.

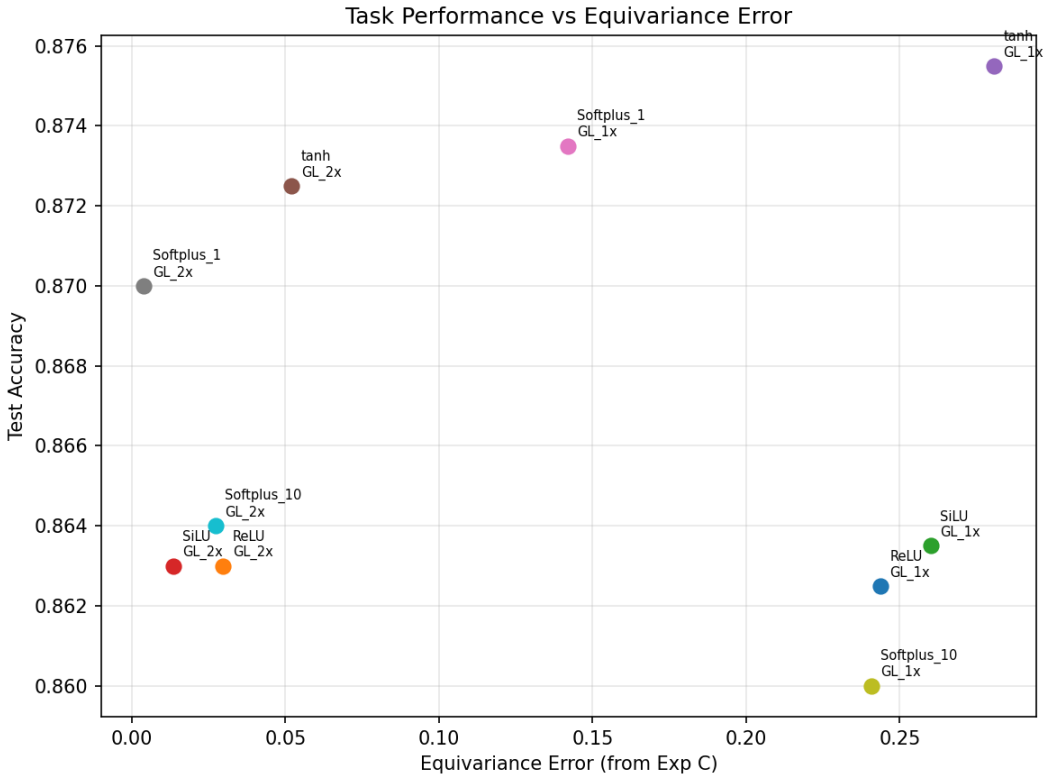

Figure 11: Synthetic task accuracy vs. equivariance error. No clear monotonic trend.

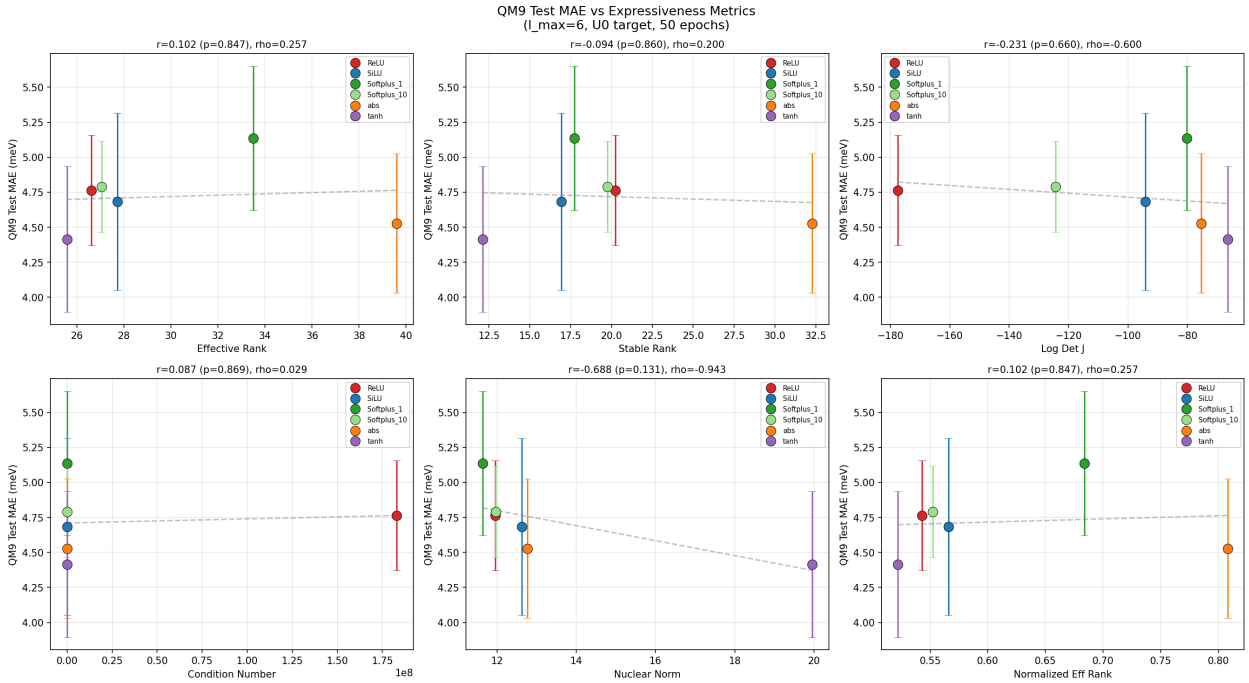

Figure 12: Downstream QM9 MAE vs. each Jacobian metric. Nuclear norm and Frobenius norm are the strongest predictors; effective rank and condition number show no significant correlation.